



ECON6025 Data Sampling | Group G1
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Project

Customer Satisfaction and Revisit Intention: Koufu, SMU

Research Area: Systematic Sampling

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Executive Summary

This study examines customer demand at the Koufu hawker centre on the SMU campus, identifying who dines there, what draws them in, and what brings them back. A systematic sample of 159 customers was surveyed, and population proportions and means were estimated with 95% bounds of error and tested using difference-of-proportions and chi-square methods.

Four hypotheses were tested. Demand is majority captive: 66.7% of customers are SMU students or faculty against 33.3% from off campus (H3). Convenience is the dominant pull driver, cited by 54.3% as their main reason for choosing Koufu, far ahead of price at 20.1% (H1). Retention, however, is led by price/value (32.7%) and food quality (23.3%), together accounting for 56.0% of stated return drivers (H2), and the main return driver differs significantly by age group (H4). Overall, 71.1% of customers were satisfied and 79.2% would recommend Koufu. The findings describe a largely captive but contestable base whose repeat demand is price- and quality-sensitive; the report therefore recommends that operators safeguard loyalty through price-value and food-quality positioning rather than relying on the campus location advantage alone.

1. Introduction

1.1 Background

Koufu operates as a major food court chain, with a presence of over 70 outlets across the north, south, east, west and central regions of Singapore (Koufu, n.d.), one of which is within the Singapore Management University (SMU) campus, in the basement of Li Ka Shing Library. This multi-stall food hall is one of several dining options within SMU, facing competition from nearby food outlets such as Food Dynasty hawker centre, Open House, and Subway in the surrounding Bras Basah area. Demand comes from a mix of students, faculty, nearby workers and tourists or residents. To measure this demand, we have used customer satisfaction and revisit intention as a proxy. Given its central location, proximity to a few nearby companies and positional advantage within the SMU campus, there is repeated demand from the customers, that determines the stability of Koufu's revenue stream from this food hall.

1.2 Problem Statement

Koufu is at the middle-lower end of the market, less competitive than the food courts like Food Republic or Food Junction (Commoncog, 2022). Due to the many food dining options near to Koufu, SMU combined with its weaker market position makes customer retention a real concern for the outlet. Customer satisfaction has been found to be a significant predictor of revisit intention in food service settings (Rajput & Gahfoor, 2020) and specifically in food court contexts (Dabral et al., 2025). Therefore, this study aims to provide a reliable estimate on which Koufu can rely on to improve customer retention and improve the revenue generated on a regular basis.

Any feedback collected informally through complaints, reviews, or comments online is voluntary, and tends to be received disproportionately from customers with strong opinions, while customers with average experiences stay silent. Such responses are subject to non-response bias and self-selection. A statistic drawn this way has no known sampling properties, so no valid conclusion can be drawn about the broader customer base from it, regardless of how many responses are collected. Acting on biased feedback risks misallocating resources toward problems that are not representative of the typical customer experience. As our study relies on a probability-based survey sampling method, it would help the relevant authorities to make sound decisions on whether to invest in service improvements, adjust pricing, or address recurring complaints systematically.

1.3 Project Objectives

This study tests four hypotheses concerning customer demand at Koufu, SMU: what attracts customers to visit, what brings them back, who makes up the customer base, and whether the retention story differs across age group.

Hypothesis 1 – Pull drivers

What attracts customers to Koufu in the first place?

H_0 : No single reason dominates the choice to eat at Koufu today.

H_1 : Locational convenience is the dominant pull driver.

Identifying the main pull factor matters for understanding whether customers are choosing Koufu deliberately, or simply because it is the most accessible option at the time. The two scenarios call for different strategies: one centred on food quality and the other on retention once customers are already through the door.

Hypothesis 2 – Retention drivers

What brings customers back?

H_0 : No attribute leads the decision to return.

H_1 : Price and food quality are the leading retention drivers.

Identifying what brings customers back is important because the reasons for a first visit and a repeat visit are not necessarily the same. If the two sets of drivers diverge, the factors that bring customers in may not be the same factors that keep them coming back.

Hypothesis 3 - Off-campus share

Who makes up Koufu's customer base?

H_0 : The off-campus share is at least 0.50 (demand is not majority on-campus).

H_1 : The off-campus share is below 0.50 (demand is majority on-campus).

A customer base made up mostly of students and faculty behaves differently from one with customers who are not part of SMU. Customers drawn in by convenience are not the same as those who actively choose Koufu, and this balance shapes how much retention effort is needed.

Hypothesis 4 - Heterogeneity in retention drivers

Does the customer retention differ across the age group?

H_0 : The retention driven profile is the same for younger (35 years old and below) and older customers (above 35 years).

H_1 : The retention driven profile differs by age group.

If all customers are drawn back by the same factors, one retention strategy is enough. If retention drivers differ by age, a single strategy may miss what one group values, making targeted improvements more effective than a blanket approach. Gender was also checked as a secondary comparison; it showed no significant difference, so it is not treated as a formal hypothesis in this study.

Together, these hypotheses show what drives first visits versus repeat visits, how concentrated Koufu's demand base is, and whether retention strategies should be tailored by age group.

The remainder of this report details the survey methodology, presents the results of these hypothesis tests, and discusses their implications for Koufu SMU.

2. Methods

2.1 Scope

The scope of this study is to investigate the composition of the customers in Koufu by age, gender, identity and incentives. We classified the customers by gender as female, male, and other; by age as group below 35 and above 35; by identity as SMU students, SMU faculties, and off-campus people, which includes nearby workers and tourists. We also classified the customers across some common incentives, including food quality and taste, price and value for money, convenient location inside SMU, familiar and comfortable, variety of food choices. Etc.

2.2 Population and Sample Size

The sampling frame for this study consists of all individuals who have visited Koufu food court at least once. Because it is not feasible to enumerate every member of this frame directly, the population was instead estimated from two components: the number of students and faculty affiliated with SMU, and the estimated density of employed individuals working within a 500-meter radius of Koufu. This estimation procedure yielded an estimated customer pool of 38,478, which is used as N throughout this study.

According to SMU's official 2025 statistical release, the university enrolled 10,294 full time undergraduates and 3,766 full time and part time postgraduates as of September 2025 (Singapore Management University, 2025). Because part time students are present on campus during class hours and therefore represent potential customers of the food court, they were not excluded from the population estimate. In addition, SMU reported 407 faculty members as of 2023 (Singapore Management University, 2023).

To estimate the population of potential off campus customers, nearby people flow was defined as the population residing or working within a 500 metre radius of Koufu. Figure 1 illustrates this catchment area.

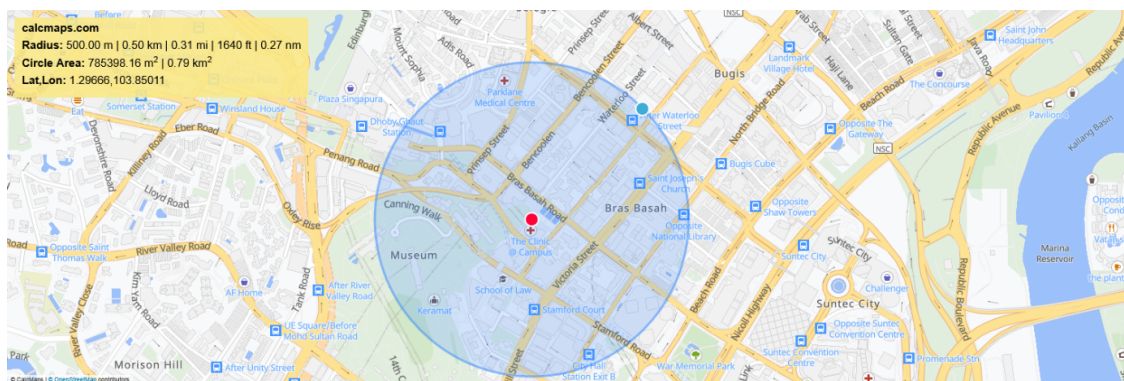


Figure 1 The Range of Nearby off-campus Area

The defined catchment area spans three planning areas under the Urban Redevelopment Authority's Master Plan 2025: Downtown Core, Museum, and Rochor (URA, 2025), with a total land area of 0.785 km². The proportion of land within the circle belonging to each planning area was then estimated from the planning area map. Approximately 25 percent of the circle's land area falls within Downtown Core, corresponding to 0.196 km². Approximately 40 percent falls within Museum, corresponding to 0.314 km². The remaining 35 percent falls within Rochor, corresponding to 0.275 km².

Population estimates for each planning area were derived from the most recent available data on Employed Residents by Planning Area of Workplace, published by the Singapore Department of Statistics, which reports the total number of employed individuals working within each planning area (Singapore Department of Statistics, 2021). To improve the precision of this estimate, the land area occupied by SMU itself was subtracted from the relevant planning areas before the calculation, since SMU's own staff and students are already accounted for separately. Approximately 70 percent of SMU's buildings lie within Museum, with the remainder situated in Rochor. The land area of these two planning areas within the 500 metre circle was therefore recalculated net of SMU's footprint.

The proportion of land within the circle relative to the total land area of each planning area was then computed, and workers were assumed to be evenly distributed across each planning area, such that the proportion of workers falling within the circle

equals the proportion of land falling within the circle. Under this assumption, the Downtown Core land within the circle represents approximately 5 percent of total Downtown Core land area, corresponding to an estimated 14,029.45 workers. The Rochor land within the circle represents approximately 16 percent of total Rochor land area, corresponding to an estimated 5,295.84 workers. The Museum land within the circle represents approximately 32.4 percent of total Museum land area, corresponding to an estimated 4,685.02 workers. Summing across the three planning areas yields an estimated 24,011 nearby off campus workers.

Although SMU is situated within the Bras Basah-Bugis district, an area with a considerable concentration of tourist attractions, daily tourist flow within the defined catchment area could not be estimated, as neither tourist arrival statistics nor public transit ridership data disaggregated to this level of geographic granularity were available. However, the number of tourists is expected to be small relative to the combined SMU and nearby worker population, and the resulting population estimate is already large enough relative to the desired level of precision that excluding tourists is unlikely to materially affect the required sample size. Given constraints on time and resources, tourists were therefore omitted from the population estimate. Combining the SMU population with the estimated nearby worker population yields an approximate total population of 38,478.

Given the time constraints of the survey period, it was not feasible to substantially reduce the margin of error. A margin of error of 0.08 was therefore adopted. Applying this margin of error to the estimated population using the standard finite population sample size formula,

$$n = \frac{Npq}{(N-1)D + pq}, \quad \text{where } D = \frac{B^2}{4}.$$

with $p = q = 0.5$ to reflect maximum variability in the absence of prior estimates (Bartlett, Kotrlik, & Higgins, 2001), yields a required sample size of 156. It is worth noting that because the estimated population is large relative to the target sample size, the finite population correction has only a marginal effect on the required sample size. Once population size exceeds roughly twenty times the uncorrected sample size, the sample size calculation effectively converges to its infinite population value, since the correction term becomes negligible (Bartlett, Kotrlik, & Higgins, 2001).

2.3 Sampling Method

To ensure that survey responses were representative of the underlying population, a 1-in-k systematic sampling method was employed to select respondents. Systematic sampling selects every kth member of the sampling frame following a randomly chosen starting point, which approximates the representativeness of simple random sampling while being substantially easier to implement in a field setting such as a food court (Etikan & Babatope, 2019). Under investigation of the sample place, the people flow during the planned sampling period is relatively low, therefore we pick 5 for value of k.

2.4 Questionnaire Design

We design a questionnaire with 10 questions asking the respondent's identity, age, gender, incentives to come and incentives to return. The full text of the questionnaire is in the appendix 1. We propose a series of hypothesis around the questionnaire to examine the customer composition of the food court.

The 5th question is designed to investigate the respondent's incentive to come to the food court. We hypothesize that customers are drawn to Koufu primarily by locational convenience rather than by the food itself, which would be verified by this question. The second hypothesis concerning retention drivers captured in Question 6, proposed that what brings customers back is value and food quality rather than the convenience that initially draws them in. The third hypothesis examined the off campus share of demand, captured in Question 1, proposing that Koufu's customer base is majority captive, meaning composed predominantly of on campus individuals. The fourth hypothesis examined heterogeneity in retention drivers across customer segments, proposing that the drivers of repeat demand differ across demographic groups. We finally match the incentive questions with the previous questions about their demographic characteristics to verify if the hypothesis are true across different groups of people.

3. Data

3.1 Survey Administration

Responses were collected over a four-day period, from 15 to 18 June, during the two peak meal periods of the day: lunch (12:00 to 1:30 PM) and dinner (6:00 to 7:30 PM). Across the four days, this produced a total of eight lunch and dinner sessions, each of which was treated as a separate data collection shift. In each shift, two to three surveyors were stationed at the tray return point (one person to each return point) to administer the questionnaire. This location was chosen deliberately, as intercepting respondents at the tray return point ensures that participants are customers who have actually completed a meal at the food court, rather than individuals who are merely present in the vicinity without having purchased food.

At the start of each shift, surveyors selected an initial respondent at random and then approached every fifth person arriving at the tray return point thereafter, consistent with standard systematic sampling procedure, in which a starting point is chosen at random and subsequent units are selected at a fixed interval (Etikan & Babatope, 2019). All responses were recorded using Microsoft Forms.

3.2 Sample Adequacy

To assess the adequacy and clarity of the questionnaire prior to full deployment, a pilot test was conducted during a single shift, yielding 33 responses. This pilot served as a preliminary check on item comprehension and response flow before the main data collection period began.

3.3 Data Cleaning

Customer incentives were classified into seven categories. To prevent incentives unrelated to the study's hypotheses from distorting the analysis, an "other" option was included for every incentive related question, allowing respondents whose motivations did not align with any of the seven predefined categories to be identified rather than forced into an inappropriate category. The same design was applied to every question in the survey with one exception: the first question, which asks respondents to identify themselves as SMU students, SMU faculty, or off campus individuals. Because these three categories are intended to be exhaustive of the entire customer population at the food court, no respondent should fall outside them, and an "other" option was therefore not needed for this item.

During data processing, responses coded as "other" were filtered out of each respective question prior to analysis. When incentive responses were cross tabulated against demographic characteristics, any respondent who selected "other" for either the demographic question or the corresponding incentive question was excluded from that particular cross tabulation, so as to avoid introducing ambiguous or non-comparable categories into the analysis.

No incomplete questionnaires were expected in the dataset, as every question except Question 9 was set as mandatory on the online survey platform. Question 9 was designed as a conditional question intended only for respondents who selected "Unlikely" or "Very unlikely" in the preceding question, which is consistent with standard survey design practice in which certain items are conditionally displayed based on a respondent's prior answers (Mirzaei, Carter, Patanwala, & Schneider, 2022). In principle, this question should therefore have applied only to a subset of respondents rather than to the full sample. In practice, however, some respondents appeared to select a response for Question 9 regardless of whether they had selected "Unlikely" or "Very unlikely" on the prior question, suggesting that the skip logic did not fully prevent unintended responses. To preserve the precision and internal consistency of the dataset, responses from individuals who selected "Likely" or "Very likely" on Question 9, despite not being routed to that item by the intended skip pattern, were filtered out of the final analysis.

4. Analysis

This section reports the findings from the survey of Koufu (SMU) customers ($n = 159$). The descriptive subsections (4.1, 4.5, 4.7) profile the respondents and the strength of the demand signals; the inferential subsections test the study's four hypotheses where each naturally falls: **H1 in 4.3 (pull drivers), H2 in 4.4 (retention drivers), H3 in 4.2 (off-campus share), and H4 in 4.6 (heterogeneity)**. All tests use the survey-sampling estimators: for a proportion, $V(\hat{p}) = \hat{p}\hat{q}/(n-1)$ with bound $B = 1.96\sqrt{V(\hat{p})}$; a hypothesis is rejected at the 95% level when its reference value lies outside the interval $\hat{p} \pm B$. Where two shares from the same question are compared, the variance includes the multinomial covariance term. Because the data come from a systematic 1-in-k intercept sample (detailed in the methodology section), the sample is treated as approximately simple random for estimation, the standard treatment for systematic samples where the customer flow shows no periodicity, so cluster or post-stratification estimators are not required. Binary coding: *satisfied* = score 4–5; *likely to revisit / recommend* = "Likely" or "Very likely". The one "Prefer not to say" gender response is excluded from gender comparisons only.

4.1 Respondent Profile

The sample is dominated by the on-campus population and by younger customers, consistent with a hawker centre inside a university. SMU students make up 51.6 per cent and faculty a further 15.1 per cent, so two-thirds of customers are drawn from the captive campus base; 73.0 per cent are below 35 and gender is almost evenly split. Visit frequency shows a highly engaged base: 41.5 per cent visit weekly or more and 24.5 per cent two to three times a month, so two-thirds are habitual rather than one-off visitors.

The full profile breakdown by respondent type, age, gender and visit frequency is given in Appendix Table A1, and the composition is shown graphically in Appendix Figure A4.

4.2 Off-Campus Share (Q1): Hypothesis H3

H3 concerns the composition of demand, specifically, whether Koufu's demand is majority captive (on-campus). This matters because the economic argument of the study rests on it: a majority-captive base behaves differently from an open-market one. H_0 : the off-campus share is at least 0.50 (demand is not majority on-campus); H_1 : it is below 0.50 (demand is majority on-campus). The estimated off-campus share is $\hat{p} = 0.333$ ($B = 0.074$; 95% CI 0.260–0.407). The interval lies entirely below 0.50, so H_0 is rejected: Koufu's demand is significantly majority on-campus (students and faculty are two-thirds of customers). At the same time the interval lies entirely above 0.25, so the off-campus segment is not negligible, it is a material minority of roughly one-third. The economic reading is therefore precise: demand is majority captive and low-effort, but a material, competition-exposed third must be actively retained, and it is this segment that the value and quality levers in 4.4 are most needed to hold.

4.3 Pull Drivers: Why Customers Came Today (Q5), Hypothesis H1

H1 asks what pulls customers in. H_0 : no single reason dominates the choice to eat at Koufu today; H_1 : locational convenience is the dominant pull driver. Combining “convenient location” and “already here / most convenient”, **convenience accounts for 54.7 per cent of choices ($B = 0.078$; $CI\ 0.470\text{--}0.625$)**, far ahead of price (20.1 per cent). The difference between convenience and the runner-up is $d = 0.346$ ($B = 0.123$; $CI\ 0.223\text{--}0.469$), and because the interval excludes 0, H_0 is rejected: convenience is the significantly dominant pull driver. Intrinsic product attributes, food quality (8.8 per cent) and variety (2.5 per cent), play only a minor role in the initial visit. This is the demand signature of a captive location: customers arrive because Koufu is convenient, not because it has won them on quality.

The full distribution of pull drivers (Q5) is given in Appendix Table A2.

4.4 Retention Drivers: What Brings Customers Back (Q6), Hypothesis H2

H2 asks what brings customers back. H_0 : no attribute leads the decision to return; H_1 : price/value and food quality are the leading retention drivers. The picture shifts decisively from convenience toward value and quality: **price (32.7 per cent, $CI\ 0.254\text{--}0.400$) and food quality (23.3 per cent, $CI\ 0.167\text{--}0.299$)** are the top two, together 56 per cent of stated retention motives. Food quality significantly exceeds the third driver, seating, with $d = 0.107$ ($B = 0.092$; $CI\ 0.015\text{--}0.198$), an interval excluding 0, so H_0 is rejected: price and food quality form a distinct leading tier. The contrast with the pull drivers in 4.3 is the study’s key finding, convenience, which pulled in the majority, is not cited as a reason to return at all, while food quality rises from 8.8 to 23.3 per cent and price from 20.1 to 32.7 per cent. Locational lock-in determines the first visit; repeat demand depends on the value proposition.

The full distribution of retention drivers (Q6) is given in Appendix Table A3; Figure 4.4 below contrasts the pull and retention distributions directly.

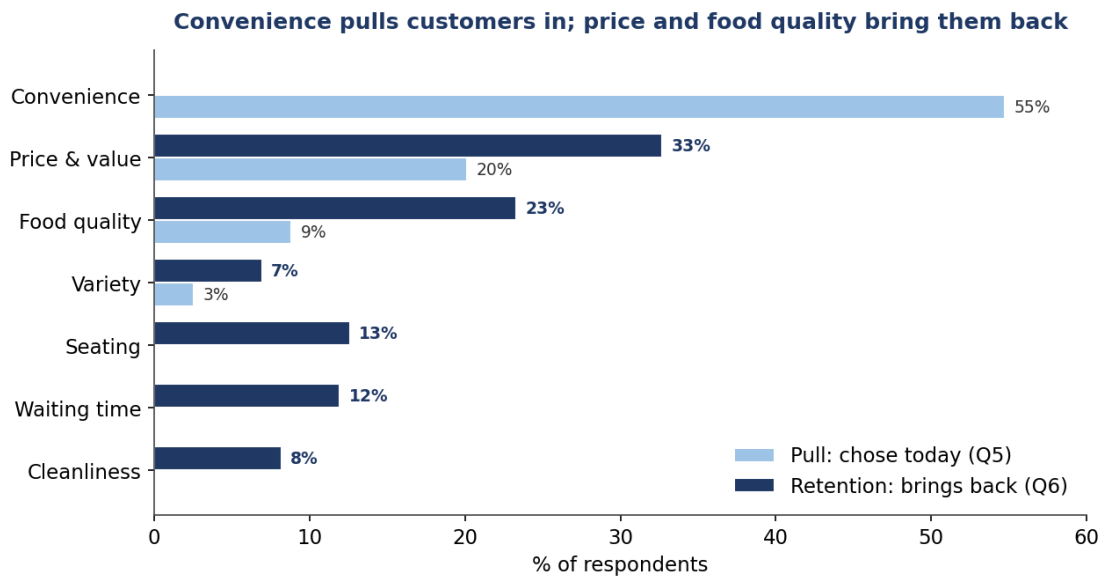


Figure 4.4 Pull (Q5) versus retention (Q6) drivers. Convenience pulls customers in but disappears as a reason to return; price and food quality rise sharply.

4.5 Repeat Demand: Revisit Intention (Q8)

Revisit intention, the primary proxy for repeat demand, is high: **83.0 per cent are likely or very likely to return within a month ($\hat{p} = 0.830$; $B = 0.059$; $CI\ 0.772\text{--}0.889$)**, with 35.8 per cent “very likely” and only 17.0 per cent at risk. This strong baseline is consistent with the captive, habitual profile in 4.1 and should be read alongside it: a high revisit rate at an on-campus centre partly reflects convenience and limited weekday substitutes rather than active loyalty. The 27 at-risk customers are examined in 4.7.

Revisit within next month (Q8)	n	%
Very likely	57	35.8%
Likely	75	47.2%
Unlikely	19	11.9%
Very unlikely	8	5.0%
Likely + Very likely (retained)	132	83.0%

Table 4.5 Revisit intention (Q8).

Repeat demand is uniform across customer segments. Breaking the revisit estimate down by respondent type (a domain estimate within the systematic sample) gives overlapping intervals for all three groups, students 0.817, faculty 0.875, off-campus 0.830, confirming that the high revisit rate is broad-based rather than concentrated in the captive student segment. Satisfaction, by contrast, is highest among off-campus customers (0.811 vs 0.646 for students), foreshadowing the segment difference in 4.6.

Respondent segment	n	Revisit $\hat{p}$ (95% CI)	Satisfied $\hat{p}$ (95% CI)
Student in SMU	82	0.817 (0.73–0.90)	0.646 (0.54–0.75)
Faculty in SMU	24	0.875 (0.74–1.00)*	0.708 (0.52–0.89)
Off-campus	53	0.830 (0.73–0.93)	0.811 (0.71–0.92)

Table 4.5b Domain estimates by respondent type (systematic sample, SRS estimators). *Faculty $n = 24$ is small; its bound is correspondingly wide.

Scaling to the population. The methodology estimates the potential customer pool at $N = 38,478$. Applying the at-risk share (unlikely to return) $\hat{p} = 0.170$ ($B = 0.059$) to this population gives an estimated $N \times \hat{p} = 38,478 \times 0.170 \approx 6,530$ customers at risk of not returning, with bound $N \times B \approx \pm 2,250$ (95% CI $\approx 4,300$ – $8,800$). This converts the revisit proportion into the expected scale of repeat-demand loss across the catchment, the quantity of direct interest to the centre, and treats the sampled visit-level proportion as applying to the potential-customer pool defined in the methodology.

The full distribution of the three outcome items is shown in Appendix Figure A5.

4.6 Heterogeneity in Demand Drivers: Hypothesis H4

H4 asks whether the demand drivers differ across demographic groups. H_0 : the share citing service and comfort attributes (seating, waiting time, cleanliness, variety) as the main reason to return is the same for younger and older customers; H_1 : it differs. Older customers cite these attributes far more often: **60.5 per cent versus 37.9 per cent for younger customers, a difference of $\hat{d} = 0.225$ ($B = 0.171$; CI 0.055–0.396)**. Because the interval excludes 0, H_0 is rejected, younger customers instead concentrate on price (36.2 per cent) and food quality (25.9 per cent). A chi-square test of independence corroborates this, with the retention-driver distribution differing significantly by age ($\chi^2 = 6.47$, $df = 2$, $p = 0.040$) though not by gender ($\chi^2 = 2.60$, $p = 0.272$). Notably, the *outcome* variables themselves do not differ by group: revisit rates are almost identical across every split (83.6% younger vs 81.4% older; 83% off-campus vs 83% on-campus), and the male–female revisit gap is not significant ($\hat{d} = 0.110$; CI -0.007 – 0.226). The heterogeneity is therefore in *what* would retain customers, not in *whether* they return, so a single retention strategy is inefficient: price and food quality are the levers for the younger majority, comfort and service attributes for the older segment. (One related result: off-campus customers are significantly more satisfied than on-campus, $\hat{d} = 0.151$; CI 0.011–0.291.)

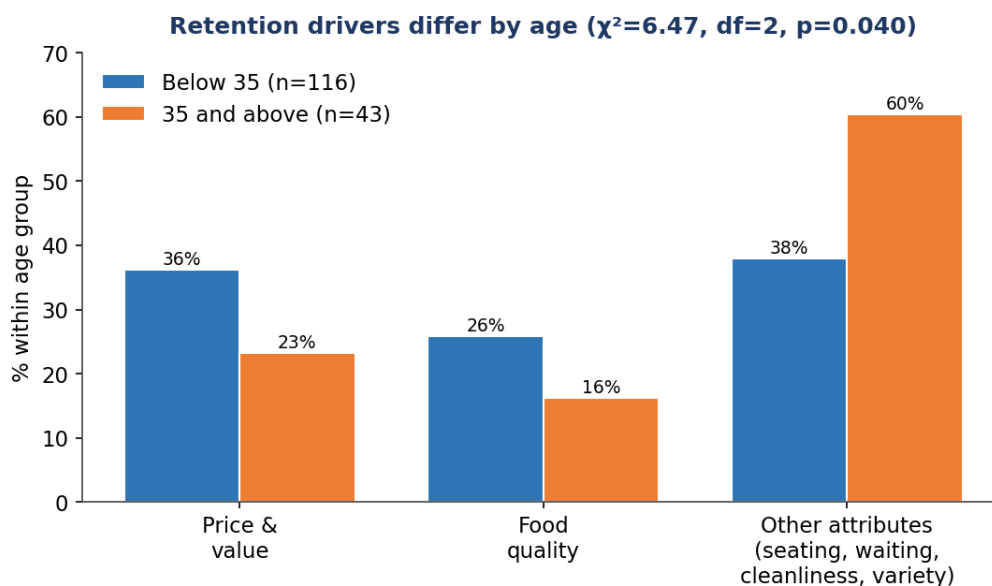


Figure 4.6 Retention drivers by age group ($\chi^2=6.47$, $df=2$, $p=0.040$).

4.7 Satisfaction and Recommendation (Q7, Q10, Q9)

Overall satisfaction (Q7) averages **3.84 out of 5 ($B = 0.135$; CI 3.70–3.97)**, with 71.1 per cent satisfied, 22.6 per cent neutral, and 6.3 per cent dissatisfied. Recommendation intention (Q10) is similar in breadth, 79.2 per cent likely to recommend,

but shallower in intensity: only 23.9 per cent are “very likely” to recommend versus 35.8 per cent “very likely” to revisit. Customers return more readily than they advocate, again reflecting convenience- rather than enthusiasm-driven demand.

Satisfaction (Q7)	n	%	Recommend (Q10)	n	%
5 Very satisfied	33	20.8%	Very likely	38	23.9%
4 Satisfied	80	50.3%	Likely	88	55.3%
3 Neutral	36	22.6%	Unlikely	27	17.0%
2 Dissatisfied	7	4.4%	Very unlikely	6	3.8%
1 Very dissatisfied	3	1.9%			
Satisfied (≥ 4)	113	71.1%	Likely + Very	126	79.2%

Table 4.7 Satisfaction (Q7, mean 3.84) and recommendation (Q10).

Among the 27 at-risk customers, the reasons for not returning (Q9) point to competition and product quality rather than price: a preference for other F&B options around SMU (8 of 27) and food quality below expectation (6) lead, with high prices and departing tourists three each. Retention failure is therefore driven by **outside options and unmet quality**, the same value and quality levers identified in 4.4. The link between satisfaction and the two downstream outcomes can be tested directly with difference-of-proportions. Satisfaction moves advocacy decisively: **91.2 per cent of satisfied customers would recommend versus 50.0 per cent of unsatisfied, $d = 0.412$ ($B = 0.154$; CI 0.258–0.565), which excludes 0**. But it does not significantly move revisit: 86.7 per cent versus 73.9 per cent gives **$d = 0.128$ ($B = 0.141$; CI –0.013–0.270), an interval that contains 0**. This asymmetry is the central economic message: convenience sustains revisit even when satisfaction is low, so satisfaction has no significant effect there, but only satisfaction converts customers into recommenders, and recommendation is what wins the off-campus demand that is not locked in by location.

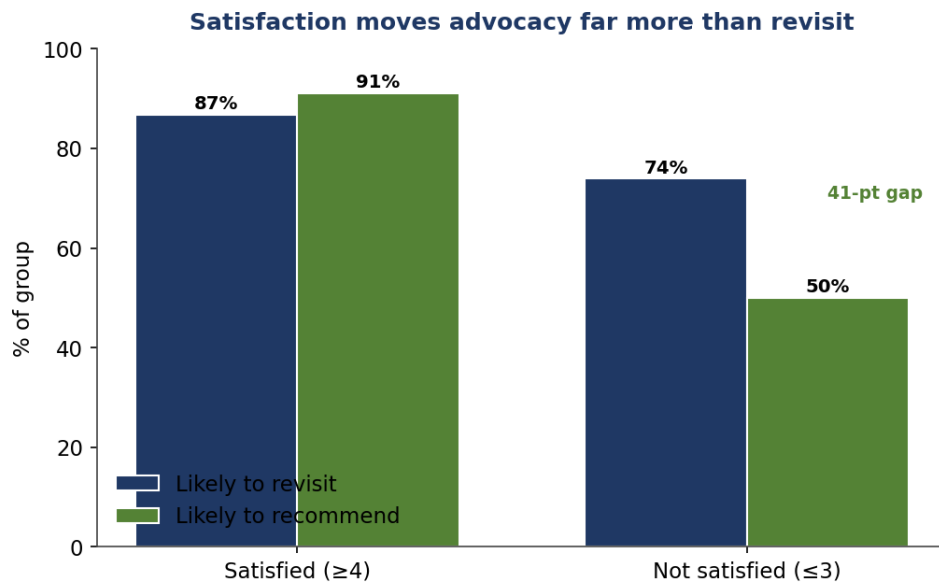


Figure 4.7 Revisit and recommendation rates by satisfaction, satisfaction moves advocacy far more than revisit.

Summary of hypothesis tests. All tests at the 5% level.

Hypothesis (H_1)	Result	Decision
H1 Convenience is the dominant pull driver (Q5)	Δ vs price 0.346 (CI 0.223–0.469)	Reject H_0
H2 Price & food quality lead retention (Q6)	Food – seating 0.107 (CI 0.015–0.198)	Reject H_0
H3 Demand is majority on-campus / captive (Q1)	Off-campus $\hat{p}$ 0.333 (CI 0.260–0.407, below 0.50)	Reject H_0
H4 Retention drivers differ by group	Service/comfort share, old–young 0.225 (CI 0.055–0.396); $\chi^2=6.47$, $p=0.040$	Reject H_0 (age)

Table 4.8 Summary of hypothesis tests.

5. Recommendations

Based on the hypotheses tested and the statistical evaluation of the survey data, several strategic interventions are proposed. Each recommendation is directly tied to the empirical findings regarding customer behavior and demand composition.

Optimising Demand Acquisition Based on Pull Drivers (Hypothesis 1): The data demonstrates that locational convenience is the significantly dominant pull driver, accounting for 54.7% of initial visits, whereas price accounts for only 20.1%. Because initial demand is overwhelmingly footfall-dependent and driven by a captive audience, Koufu should avoid allocating marketing capital toward off-campus digital advertising. Instead, management must maximize on-campus physical visibility and leverage internal Singapore Management University (SMU) student communication channels to capture the captive demographic that visits due to locational lock-in.

Prioritising Value and Quality for Customer Retention (Hypothesis 2): While convenience initiates the first visit, it does not drive repeat behavior. Rather, price and value for money (32.7%) alongside food quality (23.3%) form a distinct, leading tier of retention drivers. Therefore, Koufu must prioritize high value-for-money offerings. Introducing targeted student value meals and implementing rigorous food quality control are the most effective operational levers to convert convenience-driven, first-time visitors into a consistent, repeat customer base.

Implementing a Segmented Operational Strategy (Hypothesis 4): The analysis reveals that retention drivers differ significantly by age group ($p=0.040$), necessitating a segmented approach rather than a uniform operational strategy. Older customers (35 years and above) are heavily retained by service and comfort attributes, citing them 60.5% of the time, while younger customers (below 35 years) focus primarily on price and quality. Management should offer aggressive pricing deals to satisfy the younger student majority while simultaneously ensuring rigorous cleanliness, manageable waiting times, and comfortable seating to retain the older demographic of faculty and nearby workers.

Leveraging Satisfaction for Market Advocacy: The data indicates an asymmetry in how satisfaction influences customer behavior: convenience sustains repeat visits even when satisfaction is low, but high satisfaction decisively drives advocacy. Specifically, 91.2% of satisfied customers would recommend the food court, compared to only 50.0% of unsatisfied customers. To effectively capture the 33.3% of demand coming from the competitive off-campus market, Koufu must focus on elevating overall satisfaction beyond a “neutral” state, thereby turning existing captive customers into active promoters.

6. Limitations

The methodology and scope of this study present several distinct limitations that must be acknowledged when interpreting the inferential results and subsequent recommendations.

First, the survey relies on self-reported revisit intention as a proxy for repeat demand, with 83.0% of respondents stating they are “Likely” or “Very likely” to return. This reliance on stated preferences introduces the potential for an intention-behavior gap, wherein self-reported claims may not perfectly align with actual future footfall.

Second, the temporal scope of the data collection is heavily restricted. Because sampling was confined to brief shifts over a single-week period, the study is unable to account for longitudinal variations in customer demand, such as fluctuations across different academic terms (e.g., examination periods versus recess weeks) or differences in weekend footfall dynamics.

Third, the methodological decision to filter out “other” responses during data cleaning successfully maintained categorical consistency for cross-tabulation; however, this inherently excluded valid minority opinions and nuanced customer motivations that were not captured by the predefined survey options.

Finally, a demographic limitation exists regarding the SMU faculty segment. Because the total faculty population is inherently small, this specific cell is under-filled during the systematic sampling execution, yielding a sample size of only $n=24$. This constraint resulted in correspondingly wide confidence bounds for faculty-specific estimates, thereby reducing the statistical reliability of inferences drawn exclusively about this subgroup.

7. Conclusion

This study systematically investigated the operational dynamics and customer base composition at Koufu SMU. By employing a probability-based systematic sampling methodology, the research provides robust, empirical answers to the primary questions surrounding the food court’s demand structure.

Initial footfall is significantly dominated by locational convenience (54.7%), demonstrating that customers arrive primarily because Koufu is proximally accessible rather than choosing it as a destination for intrinsic food quality. However, the factors that pull customers in diverge from the factors that bring them back. While convenience initiates the first visit, repeat visits are distinctly driven by the combination of price/value for money and food quality, which together account for 56.0% of all stated retention motives. Provided these attributes are maintained, the baseline for repeat demand is exceptionally strong, with 83.0% of the sampled customer base indicating a high likelihood of returning within the next month.

Furthermore, the analysis confirms that Koufu’s demand is fundamentally majority on-campus. The estimated off-campus share is precisely 33.3% (95% CI: 0.260–0.407), establishing that while the captive SMU population forms the core revenue base, the off-campus segment remains a material minority exposed to market competition. Finally, the study highlights that retention drivers are not uniform across the customer base; they differ notably by age, with younger customers prioritizing economic value and food quality, whereas older customers prioritize service and comfort. Ultimately, sustaining Koufu’s

revenue stream requires a dual strategy: capitalizing on the captive audience's convenience while actively managing the distinct value and service expectations of its diverse demographic segments.

8. References

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Appendix 1: Survey Questionnaire

Customer Demand Factors in Koufu. This study investigates what draws customers to Koufu and what makes them return, with a focus on whether these retention drivers differ by age group and gender. Responses were anonymous and confidential, for study purposes only. All items were single-choice; Q1–Q8 and Q10 were required, and Q9 was conditional on a negative response to Q8.

Q1. Which best describes you?

- Student in SMU
- Faculties working in SMU
- Off-campus people: tourists or workers nearby

Q2. What is your age group?

- Below 35
- 35 and Above

Q3. What is your gender?

- Female
- Male
- Other
- Prefer not to say

Q4. How often do you visit Koufu?

- First visit today
- A few times a year
- Once a month
- 2 to 3 times a month

- Weekly or more

Q5. What was the main reason you chose to eat at this hawker centre today? (Select one)

- Food quality and taste
- Price and value for money
- Convenient location inside SMU
- Familiar and comfortable: I always come here
- Variety of food choices
- I'm already here and this was the most convenient option
- Come with others who decided
- Other

Q6. Which reason influences your decision to return most?

- Food quality and taste
- Price and value for money
- Cleanliness and hygiene
- Waiting time and speed of services
- Variety of food choices
- Seating availability and comfort
- Other

Q7. Overall, how satisfied are you with your experience today?

- 1 = Very dissatisfied
- 2 = Dissatisfied
- 3 = Neutral
- 4 = Satisfied
- 5 = Very Satisfied

Q8. How likely are you to visit this hawker centre again within the next month?

- Very likely
- Likely
- Unlikely
- Very unlikely

Q9. If you selected Unlikely or Very unlikely in Q8, what is the main reason? (Select one)

- Prices are too high
- Food quality did not meet my expectations
- Too crowded or waiting time is too long
- I prefer other food and beverage options around SMU campus
- I prefer to order food delivery instead of dining in
- I'm a tourist and will not be in Singapore
- Other

Q10. How likely are you to recommend this hawker centre to others?

- Very likely
- Likely
- Unlikely
- Very unlikely

Appendix 2: Response Summary (n = 159)

Raw response frequencies as exported from Microsoft Forms (average completion time 02:10).

Option	n
Student in SMU	82

Option	n
Faculties working in SMU	24
Off-campus people: tourists or workers nearby	53

Table B1. Which best describes you? (Q1, n = 159)

Option	n
Below 35	116
35 and Above	43

Table B2. What is your age group? (Q2, n = 159)

Option	n
Female	80
Male	78
Other	0
Prefer not to say	1

Table B3. What is your gender? (Q3, n = 159)

Option	n
First visit today	15
A few times a year	24
Once a month	15
2 to 3 times a month	39
Weekly or more	66

Table B4. How often do you visit Koufu? (Q4, n = 159)

Option	n
Food quality and taste	14
Price and value for money	32
Convenient location inside SMU	72
Familiar and comfortable: I always come here	5
Variety of food choices	4
I'm already here and this was the most convenient option	15
Come with others who decided	9
Other	8

Table B5. Main reason chosen today (Q5, n = 159)

Option	n
Food quality and taste	37
Price and value for money	52
Cleanliness and hygiene	13
Waiting time and speed of services	19
Variety of food choices	11
Seating availability and comfort	20
Other	7

Table B6. Strongest reason to return (Q6, n = 159)

Option	n
1 = Very dissatisfied	3
2 = Dissatisfied	7
3 = Neutral	36
4 = Satisfied	80
5 = Very Satisfied	33

Table B7. Overall satisfaction today (Q7, n = 159)

Option	n
Very likely	57
Likely	75
Unlikely	19
Very unlikely	8

Table B8. Likelihood of revisiting within the next month (Q8, n = 159)

Option	n
Prices are too high	11
Food quality did not meet my expectations	11
Too crowded or waiting time is too long	4
I prefer other food and beverage options around SMU campus	16
I prefer to order food delivery instead of dining in	0
I'm a tourist and will not be in Singapore	3
Other	38

Table B9. Main reason for not returning (at-risk respondents) (Q9, n = 159)

Option	n
Very likely	38
Likely	88
Unlikely	27
Very unlikely	6

Table B10. Likelihood of recommending to others (Q10, n = 159)

Appendix 3: Supporting Tables, Figures and Calculations

This appendix holds the full descriptive frequency tables and supporting figures referenced in Section 4. The hypothesis-test results in the main text are summarised visually in Figure A1.

Tables

Characteristic	Category	n	%
Respondent type (Q1)	Student in SMU	82	51.6%
	Off-campus (tourists / nearby workers)	53	33.3%
	Faculty working in SMU	24	15.1%
Age group (Q2)	Below 35	116	73.0%
	35 and above	43	27.0%
Gender (Q3)	Female	80	50.3%
	Male	78	49.1%
	Prefer not to say	1	0.6%
Visit frequency (Q4)	Weekly or more	66	41.5%
	2–3 times a month	39	24.5%

Characteristic	Category	n	%
	Once a month	15	9.4%
	A few times a year	24	15.1%
	First visit today	15	9.4%

Table A1 Respondent profile (n = 159).

Main reason chosen today (Q5)	n	%
Convenient location inside SMU	72	45.3%
Price and value for money	32	20.1%
Already here, most convenient option	15	9.4%
Food quality and taste	14	8.8%
Came with others who decided	9	5.7%
Familiar / always come here	5	3.1%
Variety of food choices	4	2.5%
Other	8	5.0%

Table A2 Pull drivers (Q5). Convenience-related reasons total 54.7%.

Strongest reason to return (Q6)	n	%
Price and value for money	52	32.7%
Food quality and taste	37	23.3%
Seating availability and comfort	20	12.6%
Waiting time and speed of service	19	11.9%
Cleanliness and hygiene	13	8.2%
Variety of food choices	11	6.9%
Other	7	4.4%

Table A3 Retention drivers (Q6).

Figures

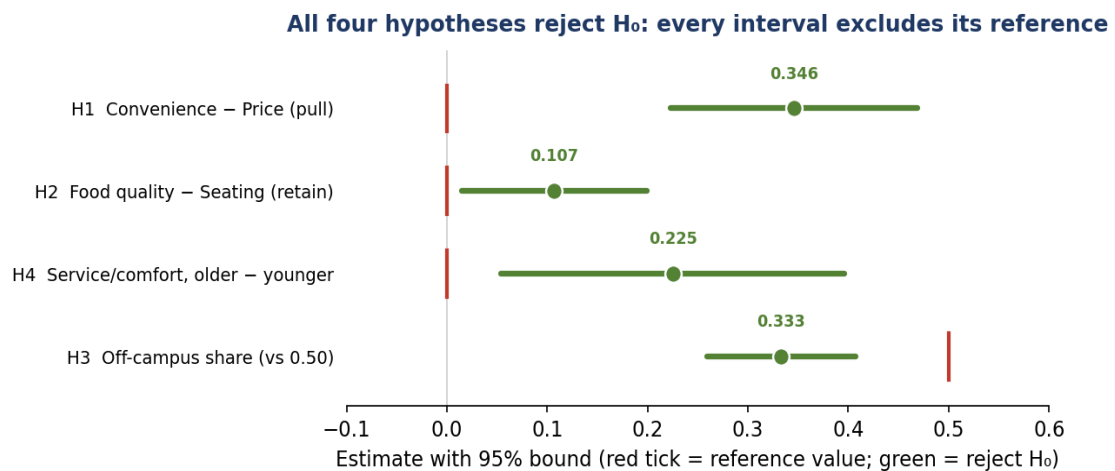


Figure A1 Hypothesis-test summary. Each bar is the point estimate $\pm$ its 95% bound; the red tick marks the reference value. All four intervals exclude their reference, so every H_0 is rejected.

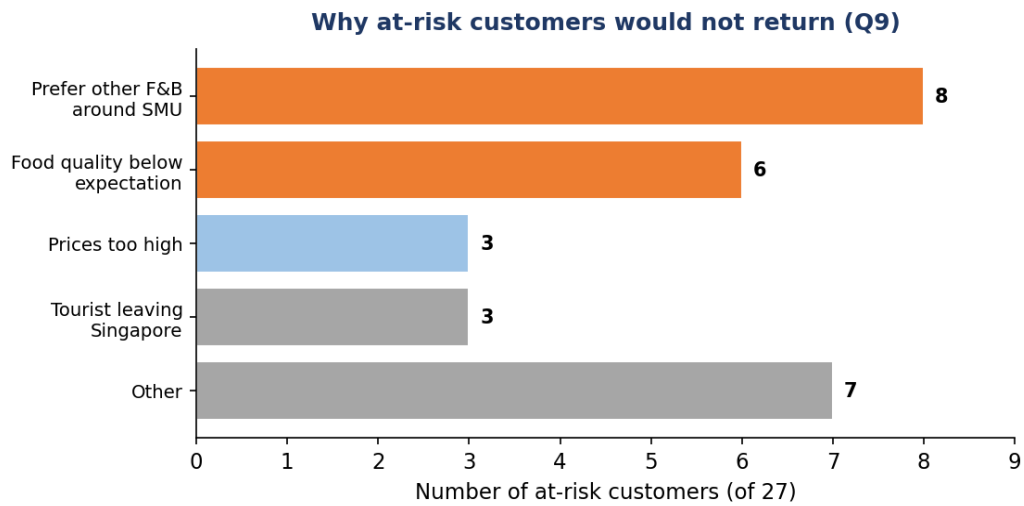


Figure A2 Reasons at-risk customers ($n = 27$) would not return (Q9). Competition from nearby F&B and unmet food quality dominate.

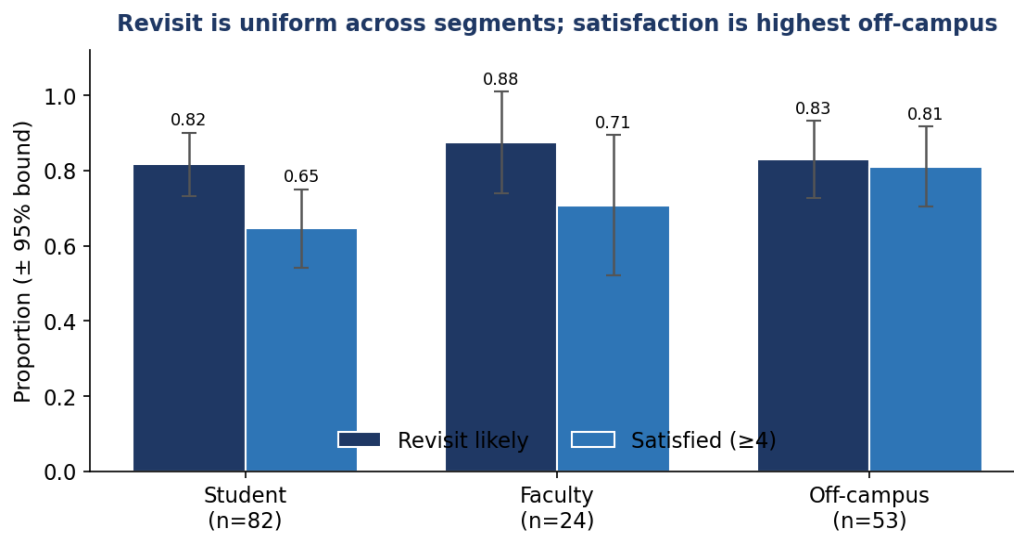


Figure A3 Domain estimates by respondent type with 95% bounds (systematic sample, SRS estimators). Revisit is uniform across segments; satisfaction is highest off-campus.

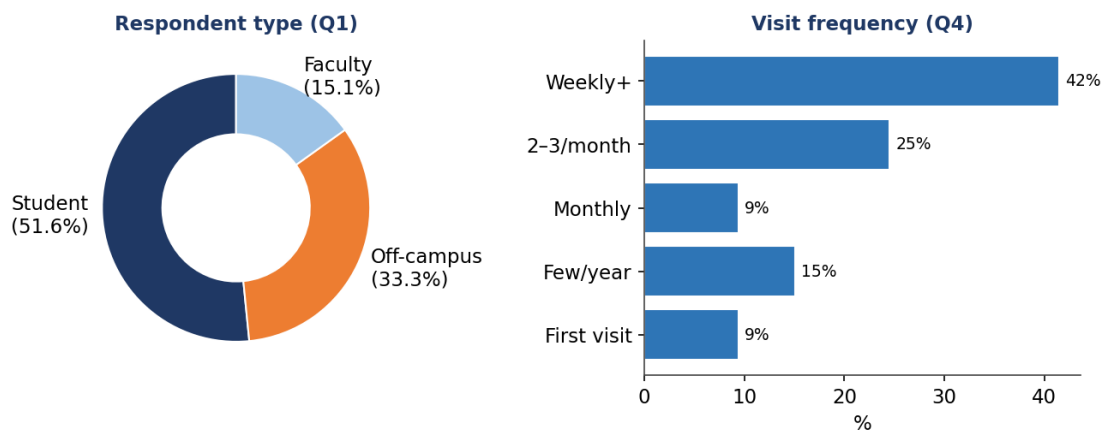


Figure A4 Respondent composition: type (Q1) and visit frequency (Q4).

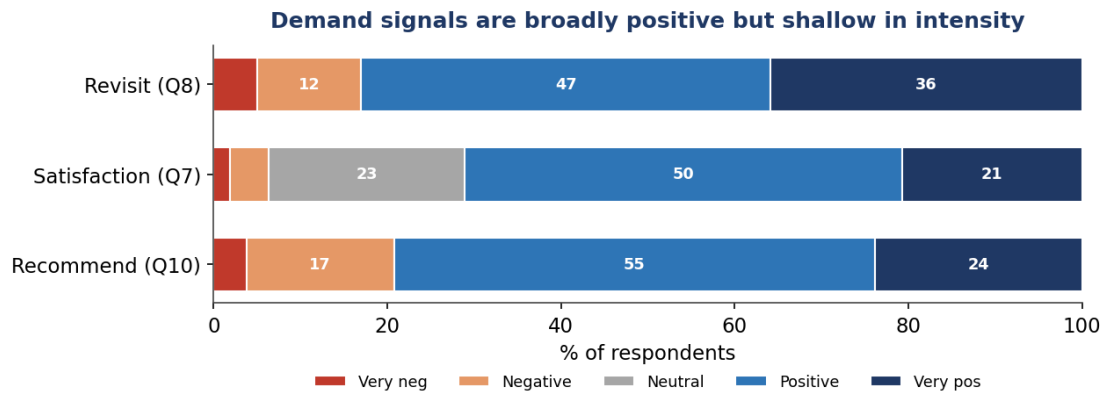


Figure A5 Distribution of the three outcome items (Q8 revisit, Q7 satisfaction, Q10 recommend), broadly positive but shallow in intensity.

Worked Calculations

Every estimate uses $B = 1.96 \cdot \sqrt{V}$. A single proportion uses $V(\hat{p}) = \hat{p}(1-\hat{p})/(n-1)$; a difference of two shares from the same question (Q5 or Q6) adds the multinomial covariance term, $V(\hat{d}) = [\hat{p}_1(1-\hat{p}_1) + \hat{p}_2(1-\hat{p}_2) + 2\hat{p}_1\hat{p}_2]/n$; a difference between two independent groups uses $V(\hat{d}) = \hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2$. Here $n = 159$.

H3, Off-campus share (Q1), one-sample proportion vs 0.50

$$\hat{p} = 53 / 159 = 0.3333$$

$$V(\hat{p}) = \hat{p}(1-\hat{p})/(n-1) = (0.3333)(0.6667)/158 = 0.001406$$

$$B = 1.96 \cdot \sqrt{0.001406} = 1.96 \times 0.03752 = 0.0735$$

$$95\% \text{ CI} = 0.3333 \pm 0.0735 = (0.2598, 0.4068)$$

Reference 0.50 lies above the upper limit $\rightarrow$ reject H_0 (demand is majority on-campus).

H1, Convenience – Price (Q5), dependent (multinomial) difference

$$\hat{p}(\text{conv}) = 87/159 = 0.5472 ; \hat{p}(\text{price}) = 32/159 = 0.2013 ; \hat{d} = 0.3459$$

$$V(\hat{d}) = [0.5472(0.4528) + 0.2013(0.7987) + 2(0.5472)(0.2013)] / 159$$

$$= [0.24778 + 0.16075 + 0.22024] / 159 = 0.62877/159 = 0.003955$$

$$B = 1.96 \cdot \sqrt{0.003955} = 1.96 \times 0.06289 = 0.1233$$

$$95\% \text{ CI} = 0.3459 \pm 0.1233 = (0.2227, 0.4692) \rightarrow \text{excludes } 0 \rightarrow \text{reject } H_0.$$

H2, Food quality – Seating (Q6), dependent (multinomial) difference

$$\hat{p}(\text{food}) = 37/159 = 0.2327 ; \hat{p}(\text{seat}) = 20/159 = 0.1258 ; \hat{d} = 0.1069$$

$$V(\hat{d}) = [0.2327(0.7673) + 0.1258(0.8742) + 2(0.2327)(0.1258)] / 159$$

$$= [0.17855 + 0.10996 + 0.05854] / 159 = 0.34705/159 = 0.002183$$

$$B = 1.96 \cdot \sqrt{0.002183} = 1.96 \times 0.04672 = 0.0916$$

$$95\% \text{ CI} = 0.1069 \pm 0.0916 = (0.0153, 0.1985) \rightarrow \text{excludes } 0 \rightarrow \text{reject } H_0.$$

H4, Service/comfort share, older – younger (Q6 $\times$ age), independent difference

$$\hat{p}(\text{old}) = 26/43 = 0.6047 ; \hat{p}(\text{young}) = 44/116 = 0.3793 ; \hat{d} = 0.2253$$

$$V(\hat{d}) = 0.6047(0.3953)/43 + 0.3793(0.6207)/116 = 0.005559 + 0.002030 = 0.007589$$

$$B = 1.96 \cdot \sqrt{0.007589} = 1.96 \times 0.08712 = 0.1707$$

$$95\% \text{ CI} = 0.2253 \pm 0.1707 = (0.0546, 0.3961) \rightarrow \text{excludes } 0 \rightarrow \text{reject } H_0.$$

Mean satisfaction (Q7), 1–5 scale

$$\bar{y} = 3.8365 ; s^2 = 0.7579$$

$$V(\bar{y}) = s^2/n = 0.7579/159 = 0.004766$$

$$B = 1.96 \cdot \sqrt{0.004766} = 1.96 \times 0.06904 = 0.1353$$

$$95\% \text{ CI} = 3.8365 \pm 0.1353 = (3.7012, 3.9718)$$

Satisfaction → outcome (4.7), independent differences (satisfied – unsatisfied)

Recommend: $\hat{p} = 103/113 = 0.9115$ vs $23/46 = 0.5000$; $d = 0.4115$

$V(d) = 0.9115(0.0885)/113 + 0.5(0.5)/46 = 0.000714 + 0.005435 = 0.006149$; $B = 0.1537$

95% CI = (0.2578, 0.5652) → excludes 0 → reject H_0 (satisfaction moves recommendation).

Revisit: $\hat{p} = 98/113 = 0.8673$ vs $34/46 = 0.7391$; $d = 0.1281$

$V(d) = 0.8673(0.1327)/113 + 0.7391(0.2609)/46 = 0.001018 + 0.004192 = 0.005210$; $B = 0.1415$

95% CI = (−0.0134, 0.2696) → contains 0 → fail to reject (convenience sustains revisit).

R Code

All estimates, bounds and tests in Section 4 are reproduced by the script below (readxl). The bound is $B = 1.96 \cdot \sqrt{V}$; comparisons of two shares from the same question use the multinomial covariance variance.

```
# =====  
# ECON6025 - Koufu @ SMU: Section 4 calculations  
# Survey-sampling estimators with bound  $B = 1.96 \cdot \sqrt{V}$   
# A hypothesis is rejected at the 95% level when its reference value  
# lies outside the interval (estimate  $\pm B$ ).  
# =====  
library(readxl)  
# ---- Load data ----  
path <- "/Users/Maria/Downloads/Custom_Demand_Factors_in_Koufu1159.xlsx"  
d <- read_excel(path)  
n <- nrow(d) # n = 159  
z <- 1.96  
# Short column codes (Excel columns 7-16; adjust if your export differs)  
names(d)[7:16] <- c("type", "age", "gender", "freq", "pull",  
"retain", "satis", "revisit", "noreturn", "recommend")  
# ---- Helper functions ----  
# Proportion estimate with bound and 95% CI (SRS estimator; the  
# systematic 1-in-k sample is treated as approximately SRS).  
prop_est <- function(x, m, label, ref = NA) {  
  p <- x / m; V <- p * (1 - p) / (m - 1); B <- z * sqrt(V)  
  cat(sprintf("%-34s p.hat=%+.4f B=%+.4f CI=(%.4f, %.4f)",  
    label, p, B, p - B, p + B))  
  if (!is.na(ref)) {  
    dec <- if (ref < p - B || ref > p + B) "excludes ref -> reject H0"  
    else "contains ref -> fail"  
    cat(sprintf(" [vs %.2f: %s]", ref, dec))  
  }  
  cat("\n"); invisible(p)  
}  
# Difference of proportions, INDEPENDENT groups:  $V = p_1q_1/n_1 + p_2q_2/n_2$   
diff_indep <- function(x1, n1, x2, n2, label) {  
  p1 <- x1 / n1; p2 <- x2 / n2  
  V <- p1 * (1 - p1) / n1 + p2 * (1 - p2) / n2; B <- z * sqrt(V); dd <- p1 - p2  
  dec <- if (dd - B > 0 || dd + B < 0) "excludes 0 -> reject H0"  
  else "contains 0 -> fail"  
  cat(sprintf("%-34s d.hat=%+.4f B=%+.4f CI=(%.4f, %.4f) %s\n",  
    label, dd, B, dd - B, dd + B, dec)); invisible(dd)  
}  
# Difference of two shares from the SAME question (multinomial,  
# dependent):  $V = [p_i(1-p_i) + p_j(1-p_j) + 2 \cdot p_i \cdot p_j] / n$   
diff_multinom <- function(xi, xj, m, label) {  
  pi <- xi / m; pj <- xj / m  
  V <- (pi * (1 - pi) + pj * (1 - pj) + 2 * pi * pj) / m  
  B <- z * sqrt(V); dd <- pi - pj  
  dec <- if (dd - B > 0 || dd + B < 0) "excludes 0 -> reject H0"  
  else "contains 0 -> fail"  
  cat(sprintf("%-34s d.hat=%+.4f B=%+.4f CI=(%.4f, %.4f) %s\n",  
    label, dd, B, dd - B, dd + B, dec)); invisible(dd)  
}  
# Mean estimate with bound:  $V = s^2 / n$   
mean_est <- function(y, label) {  
  ybar <- mean(y); s2 <- var(y); V <- s2 / length(y); B <- z * sqrt(V)  
  cat(sprintf("%-34s y.bar=%+.4f s2=%+.4f B=%+.4f CI=(%.4f, %.4f)\n",  
    label, ybar, s2, B, ybar - B, ybar + B))  
}  
# ---- Binary recodes ----  
revisit_bin <- d$revisit %in% c("Likely", "Very likely")  
satis_bin <- d$satis %in% c("4 = Satisfied", "5 = Very Satisfied")  
rec_bin <- d$recommend %in% c("Likely", "Very likely")  
offcampus <- d$type == "Off-campus people: tourists or workers nearby"
```

```

# ---- 4.2 H3: demand is majority on-campus (Q1) -----
cat("=== H3 Off-campus share (Q1) ===\n")
prop_est(sum(offcampus), n, "Off-campus share", ref = 0.50)
# ---- 4.3 H1: convenience is the dominant pull driver (Q5) -----
cat("\n=== H1 Pull drivers (Q5) ===\n")
conv <- sum(d$pull %in% c("Convenient location inside SMU",
"I'm already here and this was the most convenient option"))
price5 <- sum(d$pull == "Price and value for money")
prop_est(conv, n, "Convenience (pull)")
diff_multinom(conv, price5, n, "H1: Convenience - Price (pull)")
# ---- 4.4 H2: price & food quality lead retention (Q6) -----
cat("\n=== H2 Retention drivers (Q6) ===\n")
pr6 <- sum(d$retain == "Price and value for money")
fq6 <- sum(d$retain == "Food quality and taste")
se6 <- sum(d$retain == "Seating availability and comfort")
prop_est(pr6, n, "Price (retain, top)")
prop_est(fq6, n, "Food quality (retain, 2nd)")
diff_multinom(fq6, se6, n, "H2: Food quality - Seating")
# ---- 4.5 Repeat demand (Q8) + domain estimates by type -----
cat("\n=== 4.5 Repeat demand (Q8) ===\n")
prop_est(sum(revisit_bin), n, "Likely to revisit")
for (t in c("Student in SMU", "Faculties working in SMU",
"Off-campus people: tourists or workers nearby")) {
msk <- d$type == t
prop_est(sum(revisit_bin[msk]), sum(msk),
paste(" revisit:", substr(t, 1, 18)))
}
# ---- 4.6 H4: retention drivers differ by group -----
cat("\n=== H4 Heterogeneity (retention driver x age) ===\n")
svc <- d$retain %in% c("Seating availability and comfort",
"Waiting time and speed of services",
"Cleanliness and hygiene", "Variety of food choices", "other")
young <- d$age == "Below 35"; old <- d$age == "35 and Above"
diff_indep(sum(svc[old]), sum(old), sum(svc[young]), sum(young),
"H4: service share, old - young")
# Chi-square corroboration (price / food / other) x age
ret3 <- ifelse(d$retain == "Price and value for money", "Price",
ifelse(d$retain == "Food quality and taste", "Food", "Other"))
print(chisq.test(table(ret3, d$age), correct = FALSE))
# ---- 4.7 Satisfaction, recommendation, satisfaction -> outcome -----
cat("\n=== 4.7 Satisfaction & recommendation ===\n")
mean_est(as.numeric(substr(d$satis, 1, 1)), "Mean satisfaction (Q7)")
prop_est(sum(sat_bin), n, "Satisfied >=4 (Q7)")
prop_est(sum(rec_bin), n, "Likely to recommend (Q10)")
diff_indep(sum(rec_bin[sat_bin]), sum(sat_bin),
sum(rec_bin[!sat_bin]), sum(!sat_bin),
"Recommend: satisfied - unsat")
diff_indep(sum(revisit_bin[sat_bin]), sum(sat_bin),
sum(revisit_bin[!sat_bin]), sum(!sat_bin),
"Revisit: satisfied - unsat")

```

R Console Output

```

=== H3 Off-campus share (Q1) ===
Off-campus share p.hat=0.3333 B=0.0735 CI=(0.2598, 0.4068) [vs 0.50: excludes ref -> reject H0]
=== H1 Pull drivers (Q5) ===
Convenience (pull) p.hat=0.5472 B=0.0776 CI=(0.4696, 0.6248)
H1: Convenience - Price (pull) d.hat=+0.3459 B=0.1233 CI=(+0.2227, +0.4692) excludes 0 -> reject H0
=== H2 Retention drivers (Q6) ===
Price (retain, top) p.hat=0.3270 B=0.0732 CI=(0.2539, 0.4002)
Food quality (retain, 2nd) p.hat=0.2327 B=0.0659 CI=(0.1668, 0.2986)
H2: Food quality - Seating d.hat=+0.1069 B=0.0916 CI=(+0.0153, +0.1985) excludes 0 -> reject H0
=== 4.5 Repeat demand (Q8) ===
Likely to revisit p.hat=0.8302 B=0.0585 CI=(0.7716, 0.8887)
revisit: Student in SMU p.hat=0.8171 B=0.0842 CI=(0.7329, 0.9013)
revisit: Faculties working p.hat=0.8750 B=0.1352 CI=(0.7398, 1.0102)
revisit: Off-campus people: p.hat=0.8302 B=0.1021 CI=(0.7281, 0.9322)
=== H4 Heterogeneity (retention driver x age) ===
H4: service share, old - young d.hat=+0.2253 B=0.1707 CI=(+0.0546, +0.3961) excludes 0 -> reject H0
Pearson's Chi-squared test
data: table(ret3, d$age)
X-squared = 6.4653, df = 2, p-value = 0.03945
=== 4.7 Satisfaction & recommendation ===
Mean satisfaction (Q7) y.bar=3.8365 s2=0.7579 B=0.1353 CI=(3.7012, 3.9718)
Satisfied >=4 (Q7) p.hat=0.7107 B=0.0707 CI=(0.6400, 0.7814)
Likely to recommend (Q10) p.hat=0.7925 B=0.0632 CI=(0.7292, 0.8557)

```

Recommend: satisfied - unsat $d.\hat{a}=+0.4115$ $B=0.1537$ $CI=(+0.2578, +0.5652)$ excludes 0 -> reject H_0 Revisit: satisfied - unsat $d.\hat{a}=+0.1281$ $B=0.1415$ $CI=(-0.0134, +0.2696)$ contains 0 -> fail
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